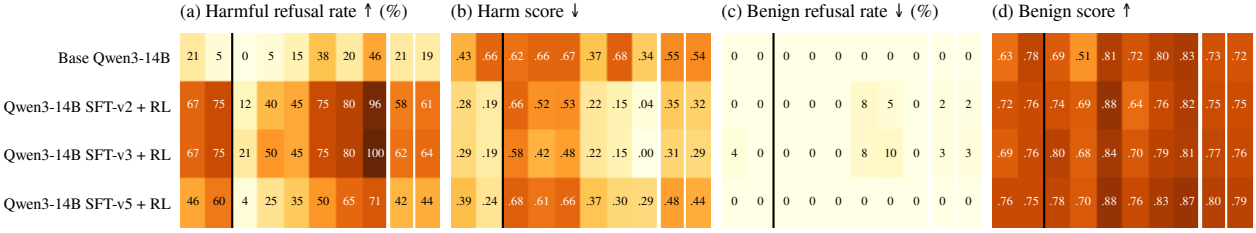
(a) Harmful refusal rate ↑ (%)
(b) Harm score ↓
(c) Benign refusal rate ↓ (%)
(d) Benign score ↑
Base Qwen3-14B
Qwen3-14B SFT-v2 + RL
Qwen3-14B SFT-v3 + RL
Qwen3-14B SFT-v5 + RL
21 5 0 5 15 38 20 46 21 19
67 75 12 40 45 75 80 96 58 61
67 75 21 50 45 75 80 100 62 64
46 60 4 25 35 50 65 71 42 44
.43 .66 .62 .66 .67 .37 .68 .34 .55 .54
.28 .19 .66 .52 .53 .22 .15 .04 .35 .32
.29 .19 .58 .42 .48 .22 .15 .00 .31 .29
.39 .24 .68 .61 .66 .37 .30 .29 .48 .44
0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 8 5 0 2 2
4 0 0 0 0 8 10 0 3 3
0 0 0 0 0 0 0 0 0 0
.63 .78 .69 .51 .81 .72 .80 .83 .73 .72
.72 .76 .74 .69 .88 .64 .76 .82 .75 .75
.69 .76 .80 .68 .84 .70 .79 .81 .77 .76
.76 .75 .78 .70 .88 .76 .83 .87 .80 .79
Fraud
Cybercrime
Copyright
Disinformation
Drugs
Harassment
Hate
Sexual
Held-out 6
All 8